The aim of this report is to survey 5 courses that are of interest to KS, and to inform him in choosing his courses in the next week. This report will aim to be objective, but take into consideration CGW's opinions on these couses and KS' priorities, which, for the CS modules, are (1) usefulness for AI safety/career, (2) time, and (3) grades in that order.

CS modules:
Maths for CSP
Probability
Deep Learning in Healthcare
Machine Learning
Algorithms and Data Structures
Models of Computation

Philosophy:
Ethics
Early Modern Philosophy
Practical Ethics/Ethics in AI
Aesthetics/Logic and Language

Y2 MT: MOC, Maths for CSP, Probability, EMP,
Y2 HT: ADS, Maths for CSP, Ethics, (AI?)
Y2 TT: (Practical Ethics?)
Y3 MT: (Quantum Information?), (Geometric Modelling?), (Logic and Language?), (Ethics of AI?)
Y3 HT: Deep Learning in Healthcare, (Physics Informed Neural Networks?), (Aesthetics?)
Y3 TT

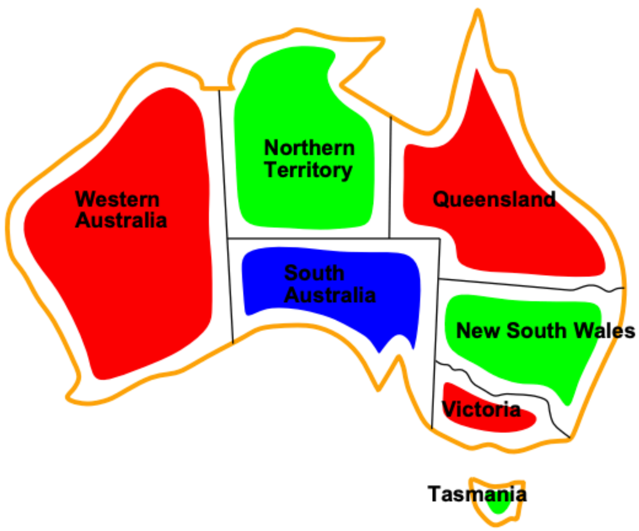
Physics informed neural networks

I recommend KS take this course. It is maths focused, very related to AI safety, and fun. However, I think this is an extremely risky decision, since the course has only been running for 1 year. Although the syllabus is more up-to-date, there might be more issues with the content being too hard, or exams being too hard/graded too harshly. It is very important that KS ask someone who has taken this course before choosing it.

Artificial intelligence

Example: Map-Coloring

Goal: assign a color to each Australian state such that no two adjacent states have the same color



Geometric modelling

Geometric modelling seems to be an almost pure maths course. It focuses on advanced Part A/B maths concepts like graphs, curves, etc.

However, personally, CGW finds this to be a bit boring, and it has almost no relationship to AI safety.

This course is best described as:

> **Applied geometry + mathematical modelling for graphics/CAD**

It sits between:

- Computational geometry
- Numerical / constructive mathematics
- Computer graphics & CAD

From the exam paper, you're dealing with:

- Solid representations (CSG vs B-Rep)
- Splines, B-splines, NURBS
- Interpolation and approximation
- Voronoi diagrams & Delaunay graphs
- Volume integrals and geometric reasoning

This is **maths-heavy**, but in a *very applied* way. The maths is mostly

- Euclidean geometry
- Polynomials & interpolation
- Integrals (nothing too wild)
- Linear algebra at a light–moderate level

2. Programming / algorithms: will you hate it?

Here's the key difference from Quantum Information.

In Geometric Modelling:

- You **won't code in exams**
- But the course *conceptually* thinks like an algorithms/graphics course
- You reason about **data structures** (winged-edge, subdivision trees)
- You think about **representations**, robustness, approximation, efficiency

Even though the exam is pen-and-paper, the *mental model* is:

> "How would this be implemented in a modelling system?"

If you *hate* algorithms because:

- You dislike runtime analysis
- You dislike coding syntax

→ this course might still be fine.

If you hate algorithms because:

- You dislike thinking about data structures
- You dislike implementation-driven reasoning

→ this course may quietly annoy you.

Quantum Information, by contrast, is much more "pure theory, no implementation intuition required".

Thus, I recommend that KS does not take Geometric Modelling.

Question 1

(a) What is a major difference between a cylinder primitive in a Constructive Solid Geometry (CSG) modeller and a cylinder primitive in a Boundary Representation (B-Rep) modeller? The difference you give should be other than the fundamental one: that the modellers represent solid and boundary, respectively. (1 mark)

(b) A cylinder has its axis aligned with the z-axis, its centroid at $(0, 0, 0)$, and is of radius 1 and height 2. What essential parameter needs to be associated with any cylinder primitive when using B-Rep with planar geometry? Describe how this cylinder is stored using a planar winged-edge B-Rep modeller. You need only describe the final data structure, rather than constructing the representation from scratch. (6 marks)

(c) By contrast with Part (b), how would a CSG modeller store the same cylinder? Note that the CSG modeller has primitives for representing planar half-spaces. (3 marks)

(d) The upper half of the model from Part (c) is dented by removing a vertical water-holding paraboloid, with an apex at $z = 0$, which touches every point on the rim at the top of the cylinder. Describe the CSG tree of this new object, called D. State the implicit functions involved in defining each component of the model. (4 marks)

(e) What is the shape of the cross-section through object D in the plane $z = 0$? What is the area A_0 of the solid part of that cross-section? What is the cross-section in $z = 1$ and its corresponding area A_1? What is the cross-section for an arbitrary $0 < z < 1$ and its corresponding area A_z? (3 marks)

(f) Using the area calculations in Part (e), calculate the volume of object D. (2 marks)

(g) Construct a formula to calculate the volume of the top part ($z \geq 0$) of object D using an integral involving the explicit function which represents the paraboloid. You should have already used a different method to calculate this volume in your answer to Part (f). Please explain how you constructed the formula, but you are not required to calculate it. (2 marks)

(h) Consider the box B with extreme vertices at $(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{4})$ and $(\frac{1}{2}, \frac{1}{2}, \frac{2}{3})$. Use interval arithmetic calculations to classify this box B against the model D as defined by your answer to Part (d). Are these interval arithmetic calculations conservative or exact? Does B lie inside D, outside D, or does it intersect the surface of D?

Suppose that a subdivision tree of boxes is being used to simplify model D by pruning the model in each box. If B were to be part of such a subdivision tree, what would the model representing object D get pruned to, in the box B? (4 marks)

Quantum information

"Quantum Computation and Quantum Information" (M.A. Nielsen and I.L. Chuang, Cambridge University Press)

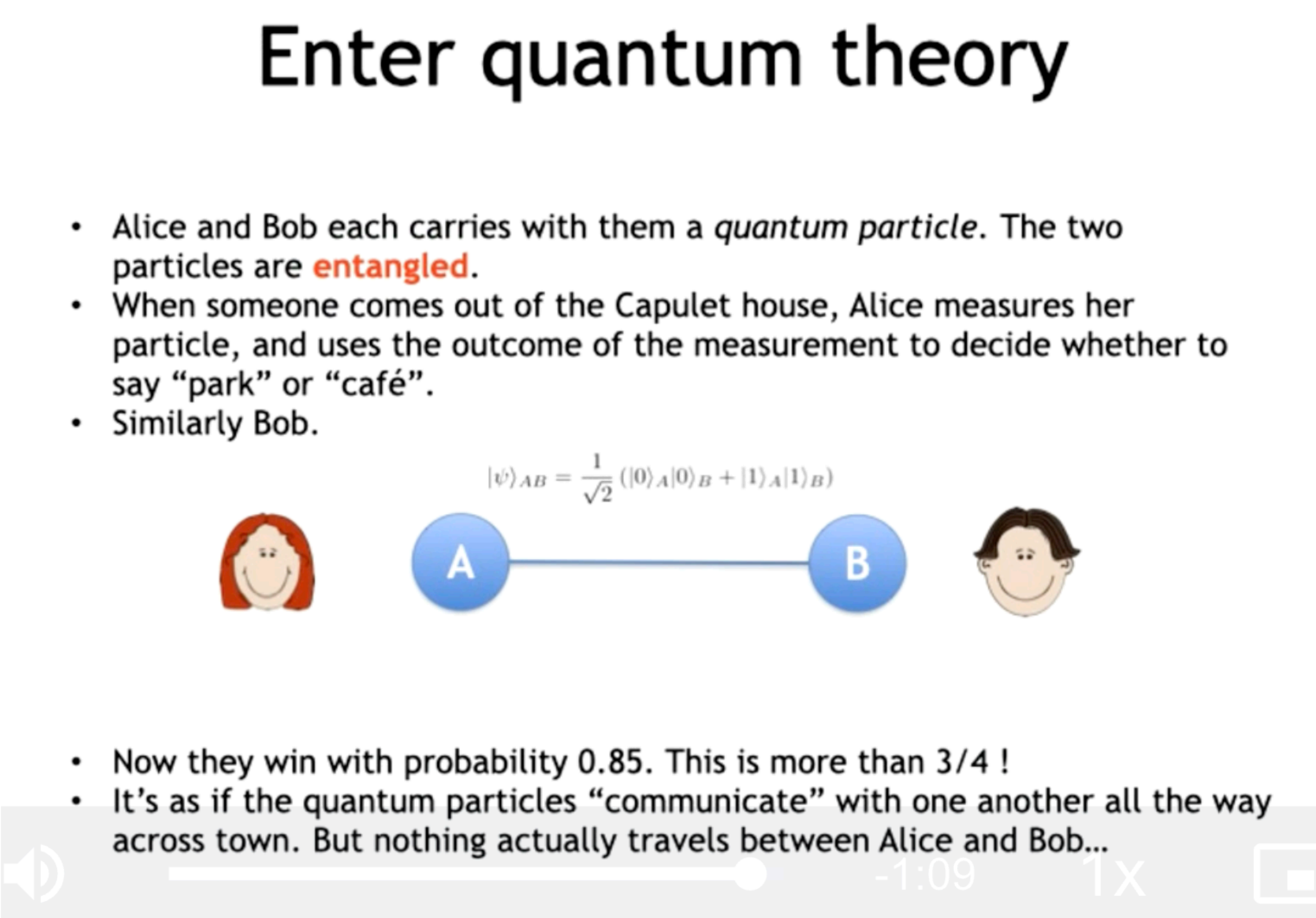
"Quantum Computer Science" (N. D. Mermin, Cambridge University Press)

I believe quantum information is the top choice for KS. I think that he would enjoy it over programming/algorithm heavy courses, since it seems very maths heavy, and that it would be good for his career, since quantum computing is a big field in computer science that could be "unlocked" by taking this course. Even if KS does not want to work in quantum computing, knowledge in this area seems very valuable, and AI safety seems to take quantum computing pretty seriously.

Some posts to read about quantum computing and AI:
https://www.lesswrong.com/posts/zrkk9ypr9S8TYiSgc/who-thinks-quantum-computing-will-be-necessary-for-ai
https://www.lesswrong.com/posts/ZkgqsyWgyDx4ZssqJ/implications-of-quantum-computing-for-artificial

However, I am unsure about the time, effort, and grades that KS might get on this course, as I am unfamiliar with the content material, and I don't know how hard the exams are.

Quantum computing seems to be a very maths and probability based course, which is verified by ChatGPT and reading the exam materials. There are no coding questions or programming type questions.



Example — the Fourier basis

The vectors: $|e_n\rangle := \frac{1}{\sqrt{d}} \sum_{m=0}^{d-1} \exp\left[\frac{2\pi imn}{d}\right] |m\rangle$

form an orthonormal basis $\{|e_n\rangle, n = 0,\ldots, d-1\}$ known as the *Fourier basis*

Very important in quantum computation!

For a qubit, $d = 2$, and we obtain

$$|e_0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) := |+\rangle$$
$$|e_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) := |-\rangle$$

Question 1

(a) Define a *basic measurement* in quantum theory. State the rule for calculating the probabilities of the outcomes and the rule that gives the state after the measurement. (4 marks)

(b) Calculate the outcome probabilities in the following cases.

(i) A basic measurement $\{|+\rangle, |-\rangle\}$ is performed on a qubit in the state

$$|\psi\rangle = \frac{i+\sqrt{2}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

(2 marks)

(ii) A basic measurement $\{|e_0\rangle, |e_1\rangle\}$ is performed on qubit A, where

$$\begin{aligned} |e_0\rangle &= \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle, \\ |e_1\rangle &= \frac{4}{5}|0\rangle - \frac{3}{5}|1\rangle, \end{aligned}$$

and qubits A and B are initially in the state

$$|\psi\rangle_{AB} = \frac{3}{5}|0\rangle_A \otimes |0\rangle_B + \frac{4}{5}|1\rangle_A \otimes |1\rangle_B.$$

(4 marks)

(c) Describe the scheme using quantum systems for unforgeable bank notes presented in lectures. Specify how the notes are designed and the test that the bank performs to check that a note is genuine. (You do not need to prove that the scheme is secure.) (5 marks)

(d) Suppose that the scheme of Part (c) is being used. A would-be forger, Eve, who knows how the scheme works, has stolen some bank details and has come to have knowledge of all the quantum states on a particular note, except for the state of the first qubit. She will attempt to forge the note, producing one copy.

(i) Suppose that Eve uses the following strategy. She performs a basic measurement, corresponding to $\{|a_0\rangle, |a_1\rangle\}$, on the first qubit. She then prepares a copy of the note, with the first qubit on the copy prepared in the state $|a_0\rangle$ or $|a_1\rangle$ according to which outcome she got. She prepares the other quantum systems on the copy in the same states as those on the original (which she can do because she knows these states). After Eve completes her procedure, both notes are tested by the bank. Suppose that

$$\begin{aligned} |a_0\rangle &= \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle, \\ |a_1\rangle &= \sin\frac{\theta}{2}|0\rangle - \cos\frac{\theta}{2}|1\rangle, \end{aligned}$$

with $0 \le \theta \le \pi$. Show that the joint probability that *both* notes pass the bank's test for authenticity is independent of θ. What is the joint probability?

Hint. Recall that $\sin 2\theta = 2\sin\theta\cos\theta$ *and* $\cos 2\theta = \cos^2\theta - \sin^2\theta$. (5 marks)

1. What *type* of course is Quantum Information?

Based on the syllabus *and* the actual exam paper you shared, this is best classified as:

> **Mathematical theoretical computer science / mathematical physics**

It sits at the intersection of:

- **Linear algebra** (vector spaces, inner products, unitary operators)
- **Probability theory** (measurement outcomes, distributions)
- **Discrete math & logic** (oracles, non-local games, no-cloning)
- **Computational complexity** (query complexity, circuit models)

What it is **not**:

- Not a software engineering course
- Not implementation-heavy
- Not algorithm-design in the "write pseudocode / prove runtime" sense you probably hate

The exam questions are proofs, calculations, and conceptual reasoning — *no coding whatsoever*. For example:

- Compute measurement probabilities
- Prove no-signalling
- Reason about entanglement and oracles
- Manipulate quantum states algebraically

That's very clearly confirmed by the exam paper itself

2025-qinfo

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If you're strong at maths and dislike programming-heavy courses, this course is **well aligned with your strengths**.

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2. Is it maths-focused enough for someone "good at maths"?

Yes — but it's a **specific flavour** of maths.

You'll mostly be doing:

- Linear algebra in complex vector spaces
- Abstract reasoning about states and transformations
- Proof-style arguments (but not epsilon–delta analysis)
- Symbolic manipulation and conceptual structure

You will *not* be doing:

- Heavy calculus
- Differential equations
- Numerical methods
- Large combinatorial algorithm proofs à la CLRS

If you like:

- Linear algebra
- Reasoning about abstract systems
- Clean axiomatic frameworks

then you'll probably enjoy it.

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3. How hard is it / grade risk?

Realistically:

- **Conceptually demanding**, but not grindy
- Difficulty comes from abstraction, not volume
- Exams reward *understanding*, not memorisation or speed-coding

The fact that candidates answer **only two questions in two hours** strongly suggests:

- Depth > breadth
- Partial progress is rewarded

- Clear reasoning matters more than technical hacks

If KS is genuinely good at maths and engages properly, this is not a “grade trap” course. It’s harder than average, but in a *fair* way.

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4. Relevance to AI safety (honest take)

This is where nuance matters.

What Quantum Information *does* help with

It builds:

- Comfort with **formal models of computation**
- Reasoning about **limits of information processing**
- Understanding of **oracles, complexity, and impossibility results**
- Exposure to LessWrong-adjacent ideas (non-local games, no-cloning, etc.)

These skills *do transfer* to AI safety research, especially theoretical safety, alignment theory, and interpretability at a deep level.

What it *doesn’t* directly give you

- ML experience
- Empirical safety skills
- Alignment-specific techniques
- Practical AI work

So: **it’s a complement, not a substitute**.

If someone did *only* quantum info and avoided all AI/ML exposure, that would be a problem. But as part of a portfolio — especially for someone maths-leaning — it’s totally defensible.

And yes, AI safety communities *do* take quantum computing seriously as a long-term consideration, even if it’s not central today.